

GaitFace: A Multimodal Dataset for Long-Range Person Identification

Alain Komaty Luis S. Luevano Vidit Vidit Zeina Al Amine
Anjith George Sébastien Marcel

Idiap Research Institute, Martigny, Switzerland

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<https://idiap.ch/paper/gaitface>

Introduction

Long-distance biometrics

- Border control faces severe congestion. Biometrics are increasingly deployed to speed up crossings. Also central to surveillance and forensic video analysis.
- Must cope with low resolution, motion blur, changing and elevated viewpoints, atmospheric effects and weather.
- **Face** and **gait** are complementary: face is highly discriminative when the resolution is good, gait could complement when the face resolution is low.

Related datasets

- In-the-wild face datasets (WebFace260M, Glint360K) are designed to improve recognition accuracy but do not reflect low-quality surveillance probes.
- Low-resolution datasets (TinyFace, SurvFace, SCFace, UCCS) lack controlled enrollment, optical-zoom captures, or temporal variability across sessions.
- Gait datasets (CASIA-B, Gait3D, CCPG, GREW) are limited to short and medium range.
- IJB-S and BRIAR are highly restricted, limiting reproducibility.

Positioning and contributions

Dataset	Dist.	Env.	#Subj.	Public
BRIAR	<1000 m	Both	~1760	No
NIST Database	10–15 m	Outdoor	30	Yes
OU-ISIR Database	~8 m	Indoor	1912	Yes
FOCS	<8.3 m	Outdoor	437	Yes
NLPR Gait	5–12 m	Outdoor	20	Yes
CASIA-B-Gait-Face	5–15 m	Indoor	124	Yes
Walking Video	<25 m	Outdoor	25	No
GaitFace	<100 m	Both	70	Yes

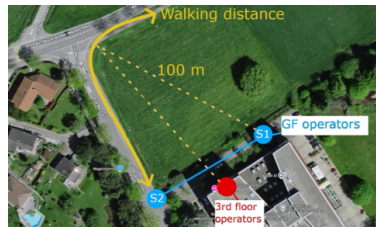
Synchronized face and gait at extreme distance is essentially absent from the public record.

Contributions

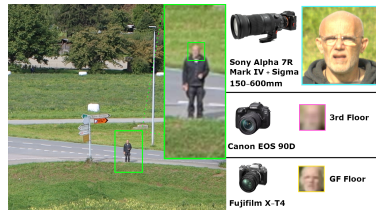
1. **GaitFace**: controlled mobile pre-enrollment, high-resolution optical-zoom face captures and long-distance face and gait.
2. **multi-session protocol** with multiple viewpoints, clothing changes and accessories.
3. **Evaluation protocols** for long-distance face, long-distance gait, and cross-session matching.
4. **Benchmarks** with state-of-the-art models, exposing the difficulty of the setting.

The GaitFace dataset

- **70 participants**, informed consent, **two sessions** weeks apart.
- **~2.7 TB** of multimodal data at distances of up to **100 m**; face and gait recorded **simultaneously**.
- Fully outdoor: sunny, cloudy, windy and rainy conditions.
- Four scenarios on a fixed trajectory: normal walking, simulated phone call, bag or backpack, different jacket.
- Two acquisition heights: **ground level** and an **elevated viewpoint** ($\approx 9\text{--}10\text{ m}$).
- Demographics recorded: 45.7% female / 54.3% male, balanced across age.



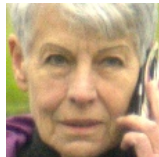
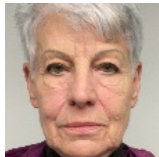
Recording positions and walking paths.



Facial resolution across cameras and heights.

Face: enrollment, probes and degradation

- **Pre-enrollment (mobile).** 15 s frontal-camera sequences on iPhone 12 and Samsung Galaxy S9. First 5 s static, 10 s of controlled head motion.
- **Outdoor face.** Subjects pause and face the sensor. A long-range zoom camera on the 3rd floor captures RAW imagery. Canon EOS 90D and Fujifilm X-T4 record simultaneously from the 3rd and ground floors.



(a) Enroll (b) Probe HQ (c) 100 m 3rd floor (d) 100 m ground floor (e) 3 m ground floor

- **Face resolution collapses with distance.** At 100 m without optical assistance: 18×22 px, IPD 4 ± 1.2 px. With optical zoom: 210×250 px, IPD 55 ± 3.5 px. At the 3 m terminal range: 450×520 px, IPD 115 ± 6.2 px.

Gait: multi-view long-distance capture

- Ground-level cameras plus the elevated third-floor viewpoint, giving frontal/lateral and top-down observations of motion.
- The third-floor setup was kept **identical** across sessions; the ground-level Fujifilm X-T4 was **deliberately relocated** between sessions, introducing a large viewpoint change.
- Four walking directions **V1–V4**; silhouettes extracted for silhouette-based models.



RGB (top) and silhouettes (bottom). Left: walking directions V1–V4. Right: normal, phone, bag, clothes.

Face recognition: protocols and models

Protocol	Enroll	Probe
HQ-[1, 3]	1, 3 (frontal)	1 (HQ)
HQ-[10, 20]	10, 20 (mix)	1 (HQ)
GF-3	3 (frontal)	[1, 10, 20, 40] (LQ)
GF-10	10 (mix)	[1, 10, 20, 40] (LQ)
3F	3 (frontal)	[1, 20, 40] (LQ)
WTC	3 (frontal)	1 (HQ/LQ)

- Border-inspired: **enroll on mobile phone data**, probe outdoors.
- HQ = 100 m with optical zoom; LQ = default lens, no optical assistance.
- Metrics follow ISO/IEC 19795: FMR against FNMR.

Model	Arch.	Params	Target
IR50	ResNet	43.6M	Robustness
IR101	ResNet	65.2M	High accuracy
EdgeFace	Hybrid	18.2M	Edge / IoT
LV-Face	ViT	140M+	Max accuracy

- AdaFace IR50/IR101 trained on WebFace4M and WebFace12M; EdgeFace on WebFace12M; LV-Face on Glint360K.
- Spans a full computational spectrum, from lightweight hybrids to large-scale Vision Transformers.

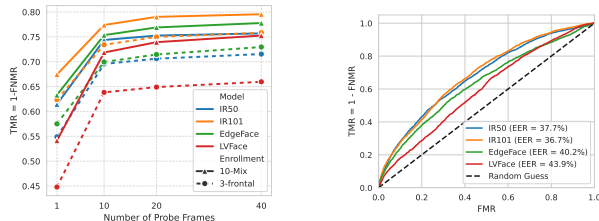
Results: HQ probes with optical zoom

Model	HQ-1		HQ-3		HQ-10		HQ-20	
	1%	0.1%	1%	0.1%	1%	0.1%	1%	0.1%
IR50	1.3	1.8	1.1	1.6	0.9	1.2	0.9	1.1
IR101	0.9	1.3	0.9	1.1	0.8	1.0	0.8	0.9
EdgeFace	1.8	2.5	1.5	2.1	1.3	1.8	1.3	1.8
LVFace	0.9	1.2	0.5	0.7	0.4	0.6	0.4	0.6

FNMR (%), thresholds calibrated on IJB-C at FMR = 1% and 0.1%.

- All four systems perform very well, with slight lead for LVFace and IR101.
- When optical zoom is used, larger models perform better.
- Adding enrollment samples helps only marginally, and **HQ-3 already matches HQ-20** $\rightarrow$ a performance cap.

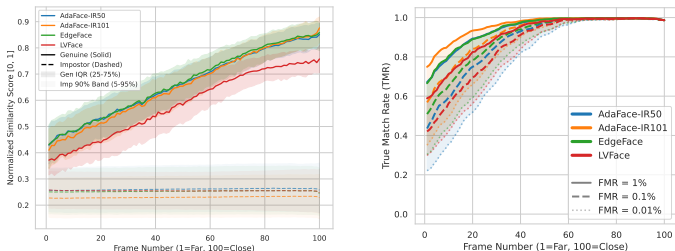
Results: LQ probes at 100 m — ground floor vs. 3rd floor



Left — **ground floor**: TMR at FMR = 1% vs. fused probe frames. Right — **3rd floor**: ROC curves under protocol 3F.

- **Ground floor**: a clear drop relative to HQ, driven purely by the loss of face resolution. More probe frames and richer enrollment help consistently, but peak TMR only reaches $\approx 80\%$.
- **EdgeFace outperforms the much heavier LV-Face**: lightweight models appear more robust to extreme resolution loss.
- **3rd floor**: elevation on top of distance causes *systemic* failure. ROC curves collapse toward chance ($AUC \approx 0.5$); EER is 36.7% (IR101), 37.7% (IR50), 40.2% (EdgeFace), 43.9% (LVFace). Standard architectures are **practically ineffective** at this range and viewpoint.

Results: walking towards the camera



Left: normalized score decay (frame 1 = far, 100 = close). Right: TMR vs. distance at three FMR thresholds.

- Genuine similarity rises monotonically as the subject approaches the camera.
- Near-perfect TMR at close range, but highly sensitive to the operating point.
- IR101 and IR50 are most resilient; EdgeFace again beats LVFace at every distance and threshold. **All models fail at the farthest frames.**

Gait recognition: protocol and baselines

Protocol

- Two camera levels: **G1** (third floor) and **G2** (ground floor, relocated between sessions → drastic viewpoint change).
- **Gallery**: one session, normal walking only — high-quality gait enrollment is unfeasible at remote registration.
- **Probe**: the entire second session — normal, phone, bag, clothes.
- Cross-view: V1 compared only against V2–V4.
- Metric: Rank-1 accuracy, closed identity set.

Baselines

- **GaitBase** (88M, GREW) — ResNet-like backbone over silhouette sequences.
- **DeepGaitV2** (95M, GREW) — novel residual layers, deeper network.
- **GaitGL** (3.1M, CASIA-B) — joint coarse global and fine local silhouette features.
- **BiggerGait** (140M, CCPG) — RGB-based, layer-wise DINOv2 features with a shallow gait encoder.

All evaluated with OpenGait using publicly released weights.

Gait results: Rank-1 accuracy (%)

Method	Train	G1 (mean over views)				G2 (mean over views)			
		Normal	Phone	Bag	Clothes	Normal	Phone	Bag	Clothes
GaitBase	GREW	26.80	21.89	23.33	24.03	21.02	18.76	19.95	15.86
DeepGaitV2	GREW	24.53	22.59	19.52	21.01	28.72	26.57	25.00	23.84
GaitGL	CASIA-B	12.25	12.50	11.07	12.08	8.46	7.05	7.07	7.37
BiggerGait	CCPG	8.78	7.67	9.64	10.67	3.18	3.00	3.55	3.86

Gallery: normal walking. Means computed excluding identical-view cases.

- Both silhouette- and RGB-based methods **generalize poorly**. No model reaches satisfactory accuracy on long-distance gait.
- G1 (small inter-session viewpoint change) is easier than G2.
- Covariates (phone, bag, clothing) degrade performance below normal walking.
- Clothing change hurts BiggerGait least (CCPG training covers large appearance changes), but its RGB pipeline suffers badly from the domain gap, which binary silhouettes largely avoid.

Conclusion and future work

- Introduced **GaitFace**, a multimodal dataset for long-range person identification built around a realistic border-crossing protocol.
- Pairing controlled high-quality mobile pre-enrollment with highly unconstrained outdoor probes exposes the operational vulnerabilities of current biometric systems.
- **Face:** state-of-the-art models degrade systemically at 100 m and from elevated viewpoints, despite remaining accurate when assisted by optical zoom.
- **Gait:** baselines struggle to generalize across viewpoint shifts, clothing variation and carried objects.
- Next: **face–gait fusion**, to improve accuracy face resolution is severely degraded by distance or atmospheric interference.
- Unconstrained long-distance biometric recognition remains an **open and highly challenging problem**.

<https://idiap.ch/paper/gaitface>

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Thank you — questions?

